

# 7.1 Abstraction ↕

## Making coffee

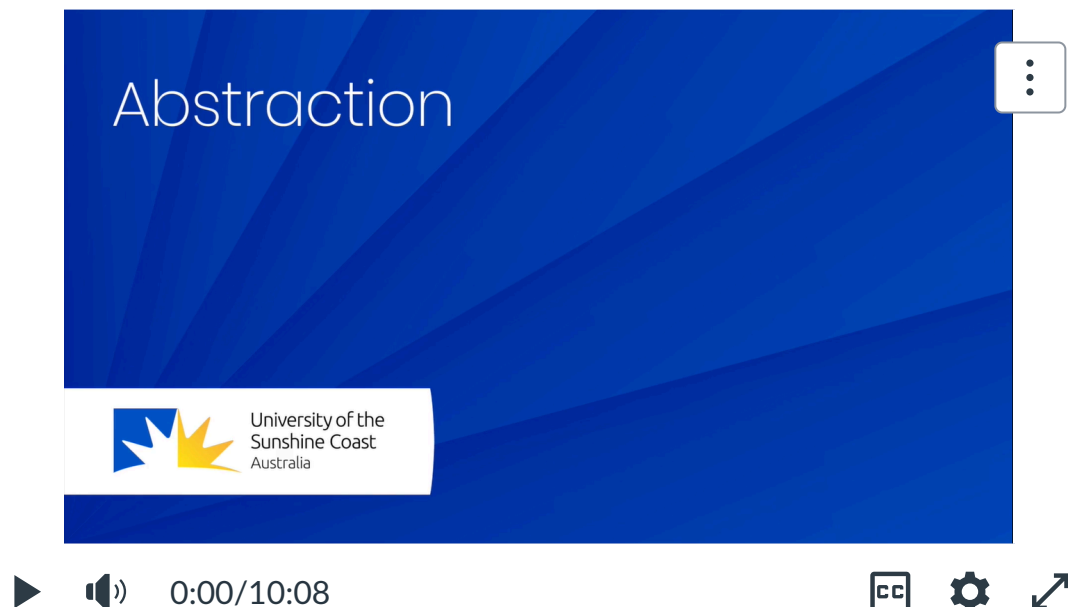
Making coffee with a coffee machine is a good example of abstraction. To make coffee, you need to know how to use your coffee machine, provide water and coffee beans, switch the machine on, select the kind of coffee you want, get milk from the fridge, clean up and a lot more. First thing in the morning, most of us do not want to focus on the details — we just want (need) a cup of coffee as fast as possible. Hence, we focus on the action — “Make coffee now” — or use language that might be even stronger!

### □ Watch: Abstraction [10.08 mins]

This video introduces abstraction in programming and contrasts abstraction with generality. This concept is used in a variety of places within programming, with functions and objects being the most-used examples.

As you watch, consider the following questions and make notes in your journal.

- As you watch, consider the following questions and make notes in your journal.
- What are some good hints and tips when trying to program with abstraction?



[Transcript \(https://learn.usc.edu.au/courses/35243/files/3137643?wrap=1\)](https://learn.usc.edu.au/courses/35243/files/3137643?wrap=1)

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## □ Files to download

Download the following slides for note taking: [ICT703 - Module 7 - Topic 1 - Abstraction.pptx \(https://learn.usc.edu.au/courses/35243/files/3137753?wrap=1\)](https://learn.usc.edu.au/courses/35243/files/3137753?wrap=1)

## □ Activity: Abstraction versus generality 1

**Abstraction ranges**  
from:

to:

**Generality ranges**  
from:

to:

**So, programming is a  
design activity where:**

- specific (does just one task)

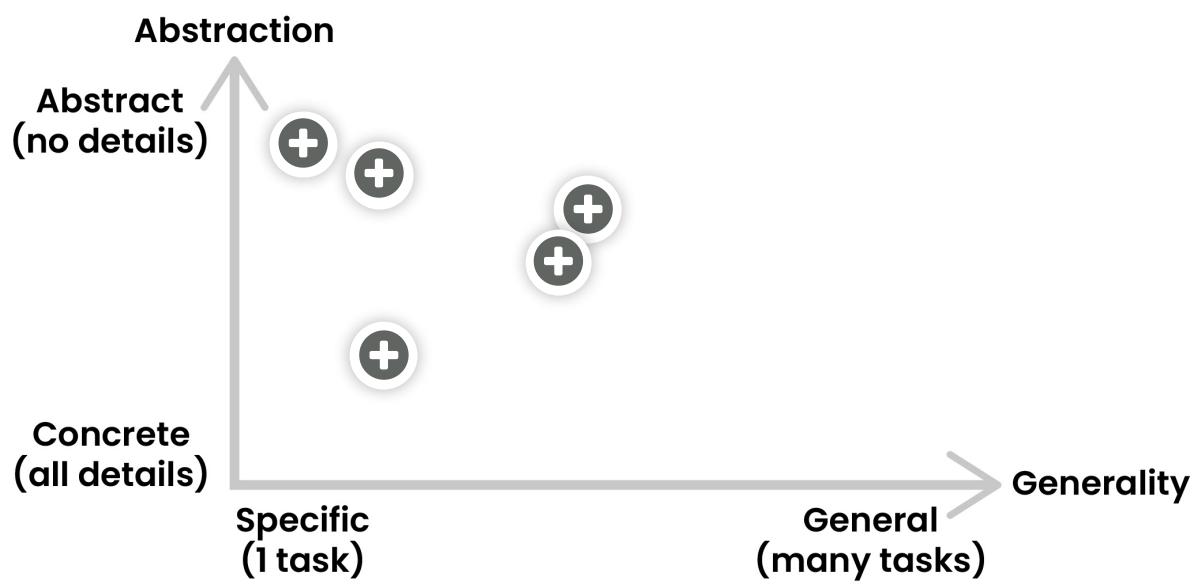
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## □ Activity: Abstraction versus generality 2

Click on the hotspots in the diagram below to see examples:



## Abstraction vs generality



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